

Name:
Neptun code:

group Z

Algorithms and data structures 1

Sample Midterm

2077. 02. 31.

Note: Every algorithm, demonstration, and general notation should be similar to, and based on everything presented during the lecture, or practice. Any method, which works fundamentally differently will be disregarded even if it produces the correct output.

Task 1. (10 points):

Sort the array below using the **insertion sort** algorithm. Demonstrate how the algorithm works in detail in every iteration.

$\langle 5, 4, 2, 8, 3 \rangle$

Task 2. (5 points):

Illustrate the **maximum selection sort** algorithm's structogram using arrays.

Task 3. (15 points):

Sort the array below with **quicksort** (the version learned in class).

a) Demonstrate how the *partition*(A, p, r) algorithm works in detail only for the first call. Let's assume that the algorithm chooses the element 14 as the pivot in the current array in the first call.

$\langle 12, 31, 23, 20, 14, 55, 11 \rangle$

b) Demonstrate how the subsequent calls are executed after the first call and write down the result of each call by showing the values in the array. Pivot can be chosen randomly.

Task 4. (15 points):

a) Provide the **polish notation** for the following infix form:

$$x := b + 2 / (a - c) - (-f \wedge b \wedge 2 * g) / z + w$$

b) Evaluate the following postfix notation. $a = 12, b = 8, c = 2, d = 3$

$$a \ b \ + \ c \ / \ d \ 2 \ \wedge \ -$$

Task 5. (15 points):

Let $H1$ and $H2$ be sorted H1Ls representing sets! Write the **Union**($H1, H2 : E1^*$) algorithm returning the union of the two sets into $H1$, by unlinking/deleting elements from $H2$. **Do not allocate new elements!** $MT(m, n) \in \Theta(m + n)$, where $|H1| = m$, and $|H2| = n$. $H1$ should remain sorted.

Algorithms and data structures 1

Sample Solutions

2077. 02. 31.

Solution 1.

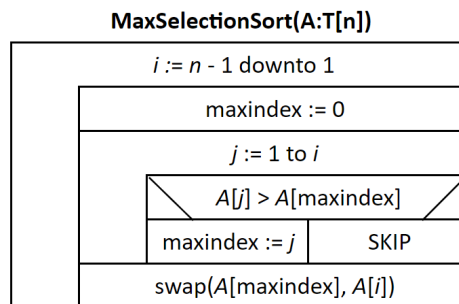
Sort the array below using the **insertion sort** algorithm. Demonstrate how the algorithm works in detail in every iteration.

$\langle 5, 4, 2, 8, 3 \rangle$

Array:	$\langle 5, 4, 2, 8, 3 \rangle$	Comments
1. iteration:	$\langle 5 \mid 4, 2, 8, 3 \rangle$	5 is sorted
2. iteration:	$\langle 4, 5 \mid 2, 8, 3 \rangle$	Inserting 4 before 5 to keep sorted
3. iteration:	$\langle 2, 4, 5 \mid 8, 3 \rangle$	...
4. iteration:	$\langle 2, 4, 5, 8 \mid 3 \rangle$	...
5. iteration:	$\langle 2, 3, 4, 5, 8 \mid \rangle$	...
Sorted array:	$\langle 2, 3, 4, 5, 8 \rangle$	

Solution 2.

Illustrate the **maximum selection sort** algorithm's structogram using arrays.



Define swap separately how it works! (using temporal variable)

Solution 3.

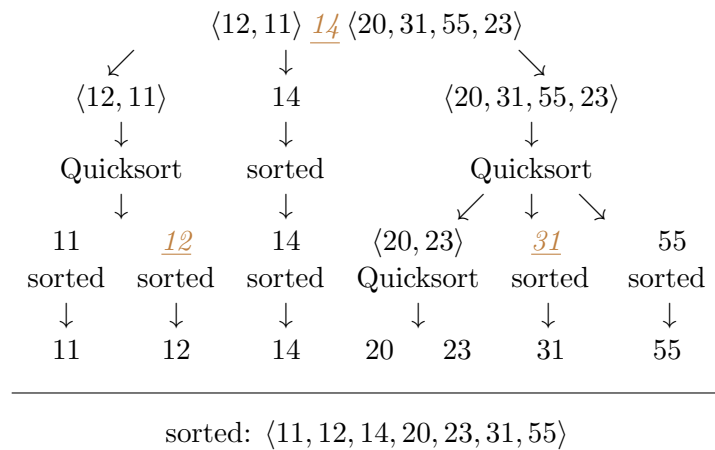
Sort the array below with **quicksort** (the version learned in class).

a) Demonstrate how the *partition*(A, p, r) algorithm works in detail only for the first call. Let's assume that the algorithm chooses the element 14 as the pivot in the current array in the first call.

$\langle 12, 31, 23, 20, 14, 55, 11 \rangle$

Array:	$\langle 12, 31, 23, 20, 14, 55, 11 \rangle$
Choosing the <i>PIVOT</i> item:	$\langle 12, 31, 23, 20, \textcolor{brown}{14}, 55, 11 \rangle$
Swap with the last element:	$\langle 12, 31, 23, 20, 11, 55, \textcolor{brown}{14} \rangle$
Look for $A[i] > \textcolor{brown}{PIVOT}$:	$\langle 12, \textcolor{blue}{31}, 23, 20, 11, 55, \textcolor{brown}{14} \rangle$
Set $j := i + 1$:	$\langle 12, \textcolor{blue}{31}, \textcolor{red}{23}, 20, 11, 55, \textcolor{brown}{14} \rangle$
Look for $A[j] < \textcolor{brown}{PIVOT}$:	$\langle 12, \textcolor{blue}{31}, 23, 20, \textcolor{red}{11}, 55, \textcolor{brown}{14} \rangle$
SWAP $A[i]$ and $A[j]$:	$\langle 12, \textcolor{blue}{11}, 23, 20, \textcolor{blue}{31}, 55, \textcolor{brown}{14} \rangle$
Set $i := i + 1$:	$\langle 12, 11, \textcolor{blue}{23}, 20, \textcolor{red}{31}, 55, \textcolor{brown}{14} \rangle$
Continue $A[j] < \textcolor{brown}{PIVOT}$:	$\langle 12, 11, \textcolor{blue}{23}, 20, \textcolor{red}{31}, 55, \textcolor{brown}{14} \rangle$
j reaches the <i>PIVOT</i> :	$\langle 12, 11, \textcolor{blue}{23}, 20, 31, 55, \textcolor{brown}{14} \rangle$
SWAP $A[i]$ with the <i>PIVOT</i> :	$\langle 12, 11, \textcolor{brown}{14}, 20, 31, 55, \textcolor{blue}{23} \rangle$
$\langle A \text{ (small)} \rangle \textcolor{brown}{PIVOT} \langle B \text{ (big)} \rangle$: $\langle 12, 11 \rangle \textcolor{brown}{14} \langle 20, 31, 55, 23 \rangle$	

b) Demonstrate how the subsequent calls are executed after the first call and write down the result of each call by showing the values in the array. Pivot can be chosen randomly.



Solution 4.

a) Provide the **polish notation** for the following infix form:

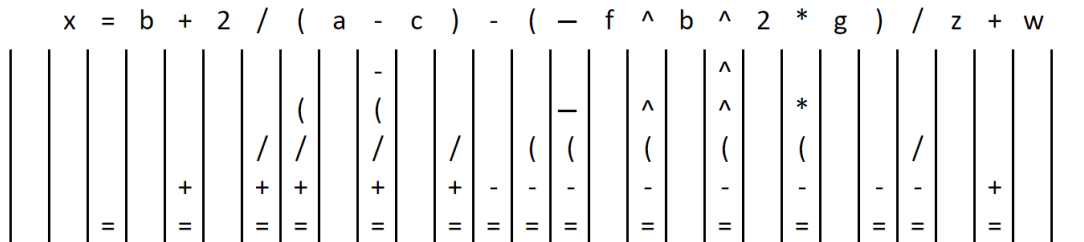
$$x := b + 2 / (a - c) - (-f \wedge b \wedge 2 * g) / z + w$$

Original expression: $x = b + 2 / (a - c) - (-f \wedge b \wedge 2 * g) / z + w$

Precedence ranking:

$\text{prec}(=) < \text{prec}(+) = \text{prec}(-) < \text{prec}(\wedge) < \text{prec}(/) < \text{prec}(\wedge) < \text{prec}(-)$

Contents of the stack: For each character whatever is currently in the stack

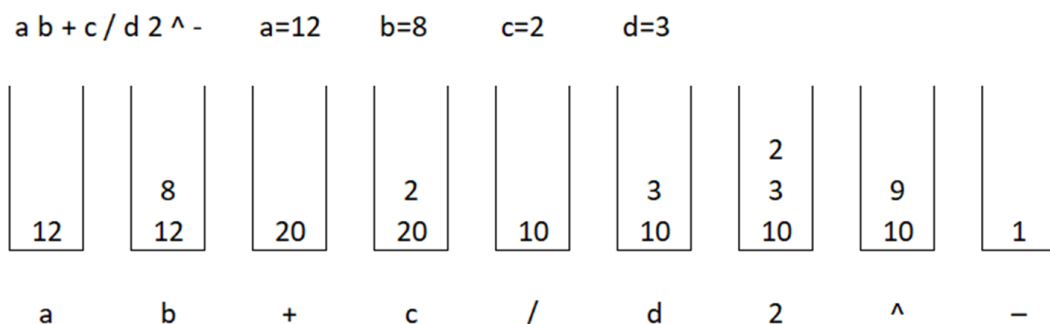


Resulting polish notation

$$x \ b \ 2 \ a \ c \ - \ / \ + \ f \ - \ b \ 2 \ ^ \ ^ \ g \ * \ z \ / \ - \ w \ + \ =$$

b) Evaluate the following postfix notation. $a = 12$, $b = 8$, $c = 2$, $d = 3$

$$a \ b \ + \ c \ / \ d \ 2 \ ^ \ -$$



Solution 5.

Let $H1$ and $H2$ be sorted H1Ls representing sets! Write the $\text{Union}(H1, H2 : E1^*)$ algorithm returning the union of the two sets into $H1$, by unlinking/deleting elements from $H2$. **Do not allocate new elements!** $MT(m, n) \in \Theta(m + n)$, where $|H1| = m$, and $|H2| = n$. $H1$ should remain sorted.

The following is an extended algorithm which does the Union with H1L and also leaves the intersection elements in the second list.

unionIntersection(Hu, Hi : E1*)			
qe:=Hu; q:=Hu->next; re:=Hi; r:=Hi->next			
q ≠ 0 ∧ r ≠ 0			
q->key < r-> key	q->key > r-> key	q->key = r-> key	
qe:=q q:=q->next	re->next:=r->next	qe:= q q:=q->next re:=r r:=r->next	
	qe->next:=r		
	r->next:=q		
	qe:=r		
	r:=re->next		
r ≠ 0			
qe->next := r		skip	
re-> next:= 0			